



Yongzheng Zhang

Tel: (+86) 13855330929

Email: yongzheng2024@outlook.com

Github: github.com/yongzheng2024

EDUCATION

ShanghaiTech University

September 2024 - Present

Master's Candidate in Computer Science and Technology

Shanghai, China

Related Courses: Algorithm Design and Analysis, Software Analysis, Computer-aided Verification

Research Interests: Network Synthesis and Verification, Programming Languages, and Network Scheduler

- Familiar with network synthesis tool, Netcomplete, source code and technical principles.
- Familiar with network verification tools, Batfish and Minesweeper, source code and technical principles.

Hainan University

September 2017 - July 2021

Bachelor of Engineering in Network Engineering

Haikou, China

Related Courses: Data Structure, Computer Architecture, Computer Operating System, Database Principles, Software Engineering, Computer Network, Routing and Switching

RESEARCH INTERNSHIP

Tsinghua University

April 2024 - September 2024

Research Intern

Beijing, China (Remote)

Worked on **ClubHeap**, a high-speed, scalable programmable packet scheduler based on the PIFO (the push-in first-out queue) framework and priority queue structure with full pipeline architecture.

- Implemented the initial design of ClubHeap, a full pipeline architecture based on PIFO and priority queue.
- Developed a software simulator to verify the hardware design and conducted experiments in Vivado.

National University of Defense Technology

July 2021 - May 2022

Assistant Engineer

Changsha, China

Worked on **ResFuz**, a fuzzing engine for estimating worst-case resource usage by resource-usage-aware fuzzing based on AFL (American Fuzzy Lop).

- Responsible for managing heap and user-defined resources, and maintaining Shell and Coccinelle scripts.
- Engaged in scientific research and submitted artifacts for FASE'22, earning available and functional badges.

WORK EXPERIENCE

HUAWEI · Data Communication

February 2023 - May 2023

Software Testing Engineer

Nanjing, China

Worked on **NetEngine SD-WAN AR6700v/AR6700v-lite**, that supported resource pooling, on-demand subscription and other cloud-native technologies to build a secure and reliable multi-tenant operation solution.

- Responsible for automated testing of layer-2 and layer-3 network protocols (control-plane protocols).
- Developed automated testing scripts using the Pytest framework, and created general check methods.
- Learned the principles of layer-2 and layer-3 network protocols, such as ARP, DNS, DHCP, VXLAN and BGP.

RESEARCH PUBLICATIONS

Localized Explanations for Automatically Synthesized Network Configurations (HotNets'24, co-first author)

TECHNICAL SKILLS

Programming Languages: Proficient in C/C++, familiar with Python, Java, P4, Shell, and Coccinelle.

Development Tools: Strong knowledge of the Linux environment, familiar with Vim, GCC, GDB, and Make.